



Celebrity Cruises' *Oasis of the Seas*, one of the world's largest and most expensive cruise liners, is shown here under construction in Turku, Finland. The 6,000-passenger ship includes a beach pool, shopping mall, and spa. Managing this project uses the same project management techniques as managing the remodeling of a store, installing a new production line, or implementing a new computer system.

Probability of Project Completion

The critical path analysis helped us determine that Milwaukee Paper's expected project completion time is 15 weeks. Julie Ann Williams knows, however, that there is significant variation in the time estimates for several activities. Variation in activities that are on the critical path can affect the overall project completion time—possibly delaying it. This is one occurrence that worries the plant manager considerably.

PERT uses the variance of critical path activities to help determine the variance of the overall project. Project variance is computed by summing variances of *critical* activities:

$$\sigma_p^2 = \text{Project variance} = \Sigma(\text{variances of activities on critical path}) \quad (3-8)$$

Example 9

COMPUTING PROJECT VARIANCE AND STANDARD DEVIATION FOR MILWAUKEE PAPER

Milwaukee Paper's managers now wish to know the project's variance and standard deviation.

APPROACH▶ Because the activities are independent, we can add the variances of the activities on the critical path and then take the square root to determine the project's standard deviation.

SOLUTION▶ From Example 8 (Table 3.4), we have the variances of all of the activities on the critical path. Specifically, we know that the variance of activity A is 0.11, variance of activity C is 0.11, variance of activity E is 1.00, variance of activity G is 1.78, and variance of activity H is 0.11.

Compute the total project variance and project standard deviation:

$$\text{Project variance } (\sigma_p^2) = 0.11 + 0.11 + 1.00 + 1.78 + 0.11 = 3.11$$

which implies:

$$\text{Project standard deviation } (\sigma_p) = \sqrt{\text{Project variance}} = \sqrt{3.11} = 1.76 \text{ weeks}$$

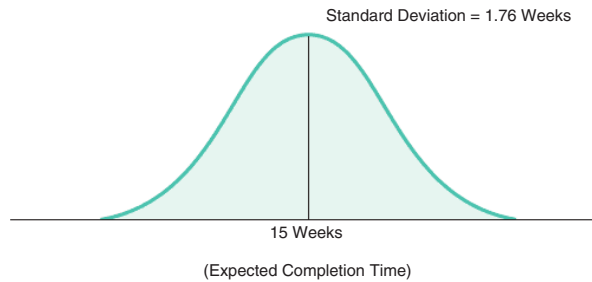
INSIGHT▶ Management now has an estimate not only of expected completion time for the project but also of the standard deviation of that estimate.

LEARNING EXERCISE▶ If the variance for activity A is actually 0.30 (instead of 0.11), what is the new project standard deviation? [Answer: 1.817.]

RELATED PROBLEMS▶ 3.17e, 3.24

Figure 3.12

Probability Distribution
for Project Completion Times
at Milwaukee Paper



How can this information be used to help answer questions regarding the probability of finishing the project on time? PERT makes two more assumptions: (1) total project completion times follow a normal probability distribution, and (2) activity times are statistically independent. With these assumptions, the bell-shaped normal curve shown in Figure 3.12 can be used to represent project completion dates. This normal curve implies that there is a 50% chance that the manufacturer's project completion time will be less than 15 weeks and a 50% chance that it will exceed 15 weeks.

Example 10

PROBABILITY OF COMPLETING A PROJECT ON TIME

Julie Ann Williams would like to find the probability that her project will be finished on or before the 16-week Earth Day deadline.

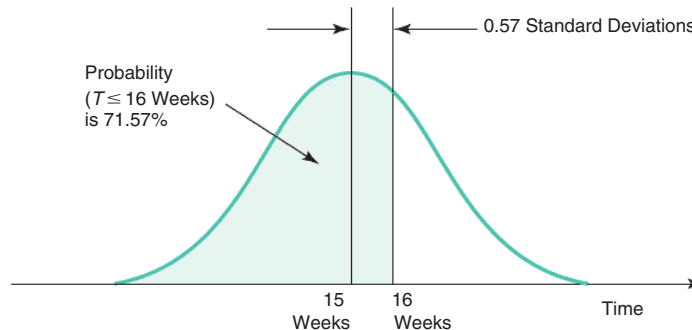
APPROACH► To do so, she needs to determine the appropriate area under the normal curve. This is the area to the left of the 16th week.

SOLUTION► The standard normal equation can be applied as follows:

$$Z = (\text{Due date} - \text{Expected date of completion}) / \sigma_p \\ = (16 \text{ weeks} - 15 \text{ weeks}) / 1.76 \text{ weeks} = 0.57 \text{ (3-9)}$$

where Z is the number of standard deviations the due date or target date lies from the mean or expected date.

Referring to the Normal Table in Appendix I (alternatively using the Excel formula `=NORMSDIST(0.57)`), we find a Z -value of 0.57 to the right of the mean indicates a probability of 0.7157. Thus, there is a 71.57% chance that the pollution control equipment can be put in place in 16 weeks or less. This is shown in Figure 3.13.



INSIGHT► The shaded area to the left of the 16th week (71.57%) represents the probability that the project will be completed in less than 16 weeks.

LEARNING EXERCISE► What is the probability that the project will be completed on or before the 17th week? [Answer: About 87.2%.]

RELATED PROBLEMS► 3.17f, 3.19d, 3.20d, 3.21b, 3.23, 3.25 (3.26e,f,g are available in [MyLab Operations Management](#))

STUDENT TIP

Here is a chance to review your statistical skills and use of a normal distribution table (Appendix I).

Figure 3.13

Probability That Milwaukee
Paper Will Meet the 16-Week
Deadline

Determining Project Completion Time for a Given Confidence Level Let's say Julie Ann Williams is worried that there is only a 71.57% chance that the pollution control equipment can be put in place in 16 weeks or less. She thinks that it may be possible to plead with the board of directors for more time. However, before she approaches the board, she wants to arm herself with sufficient information about the project. Specifically, she wants to find the deadline by which she has a 99% chance of completing the project. She hopes to use her analysis to convince the board to agree to this extended deadline, even though she is aware of the public relations damage the delay will cause.

Clearly, this due date would be greater than 16 weeks. However, what is the exact value of this new due date? To answer this question, we again use the assumption that Milwaukee Paper's project completion time follows a normal probability distribution with a mean of 15 weeks and a standard deviation of 1.76 weeks.

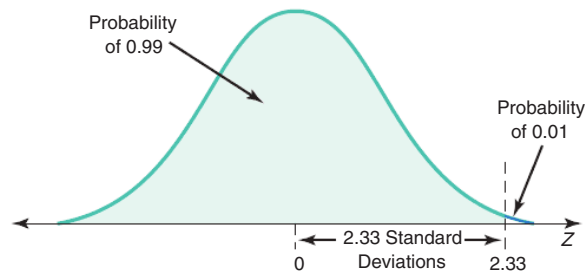
Example 11

COMPUTING PROBABILITY FOR ANY COMPLETION DATE

Julie Ann Williams wants to find the due date that gives her company's project a 99% chance of *on-time* completion.

APPROACH ▶ She first needs to compute the Z-value corresponding to 99%, as shown in Figure 3.14. Mathematically, this is similar to Example 10, except the unknown is now the due date rather than Z.

Figure 3.14
Z-Value for 99% Probability
of Project Completion at
Milwaukee Paper



SOLUTION ▶ Referring again to the Normal Table in Appendix I (alternatively using the Excel formula `=NORMSINV(0.99)`), we identify a Z-value of 2.33 as being closest to the probability of 0.99. That is, Julie Ann Williams's due date should be 2.33 standard deviations above the mean project completion time. Starting with the standard normal equation [see Equation (3-9)], we can solve for the due date and rewrite the equation as:

$$\begin{aligned}\text{Due date} &= \text{Expected completion time} + (Z \times \sigma_p) \\ &= 15 + (2.33 \times 1.76) = 19.1 \text{ weeks (3-10)}\end{aligned}$$

INSIGHT ▶ If Williams can get the board to agree to give her a new deadline of 19.1 weeks (or more), she can be 99% sure of finishing the project by that new target date.

LEARNING EXERCISE ▶ What due date gives the project a 95% chance of on-time completion? [Answer: About 17.9 weeks.]

RELATED PROBLEMS ▶ 3.21c, 3.23e

Variability in Completion Time of Noncritical Paths In our discussion so far, we have focused exclusively on the variability in the completion times of activities on the critical path. This seems logical because these activities are, by definition, the more important activities in a project network. However, when there is variability in activity times, it is important that we also investigate the variability in the completion times of activities on *noncritical* paths.

Consider, for example, activity D in Milwaukee Paper's project. Recall from Overlay 3 in Figure 3.10 (in Example 7) that this is a noncritical activity, with a slack time of 1 week. We have therefore not considered the variability in D's time in computing the probabilities of project completion times. We observe, however, that D has a variance of 0.44 (see Table 3.4 in

Example 8). In fact, the pessimistic completion time for D is 6 weeks. This means that if D ends up taking its pessimistic time to finish, the project will not finish in 15 weeks, even though D is not a critical activity.

For this reason, when we find probabilities of project completion times, it may be necessary for us to not focus only on the critical path(s). Indeed, some research has suggested that expending project resources to reduce the variability of activities not on the critical path can be an effective element in project management. We may need also to compute these probabilities for noncritical paths, especially those that have relatively large variances. It is possible for a noncritical path to have a smaller probability of completion within a due date, when compared with the critical path. Determining the variance and probability of completion for a noncritical path is done in the same manner as Examples 9 and 10.

What Project Management Has Provided So Far Project management techniques have thus far been able to provide Julie Ann Williams with several valuable pieces of management information:

1. The project's expected completion date is 15 weeks.
2. There is a 71.57% chance that the equipment will be in place within the 16-week deadline. PERT analysis can easily find the probability of finishing by any date Williams is interested in.
3. Five activities (A, C, E, G, and H) are on the critical path. If any one of these is delayed for any reason, the entire project will be delayed.
4. Three activities (B, D, F) are not critical and have some slack time built in. This means that Williams can borrow from their resources, and, if necessary, she may be able to speed up the whole project.
5. A detailed schedule of activity starting and ending dates, slack, and critical path activities has been made available (see Table 3.3 in Example 6).

Cost-Time Trade-Offs and Project Crashing

While managing a project, it is not uncommon for a project manager to be faced with either (or both) of the following situations: (1) the project is behind schedule, and (2) the scheduled project completion time has been moved forward. In either situation, some or all of the remaining activities need to be speeded up (usually by adding resources) to finish the project by the desired due date. The process by which we shorten the duration of a project in the cheapest manner possible is called project **crashing**.

Crashing

Shortening activity time in a network to reduce time on the critical path so total completion time is reduced.

CPM is a technique in which each activity has a *normal* or *standard* time that we use in our computations. Associated with this normal time is the *normal* cost of the activity. However, another time in project management is the *crash time*, which is defined as the shortest duration required to complete an activity. Associated with this crash time is the *crash cost* of the activity. Usually, we can shorten an activity by adding extra resources (e.g., equipment, people) to it. Hence, it is logical for the crash cost of an activity to be higher than its normal cost.

The amount by which an activity can be shortened (i.e., the difference between its normal time and crash time) depends on the activity in question. We may not be able to shorten some activities at all. For example, if a casting needs to be heat-treated in the furnace for 48 hours, adding more resources does not help shorten the time. In contrast, we may be able to shorten some activities significantly (e.g., frame a house in 3 days instead of 10 days by using three times as many workers).

Likewise, the cost of crashing (or shortening) an activity depends on the nature of the activity. Managers are usually interested in speeding up a project at the least additional cost. Hence, when choosing which activities to crash, and by how much, we need to ensure the following:

- ◆ The amount by which an activity is crashed is, in fact, permissible
- ◆ Taken together, the shortened activity durations will enable us to finish the project by the due date
- ◆ The total cost of crashing is as small as possible

Crashing a project involves four steps:

LO 3.6 Crash a project

STEP 1: Compute the crash cost per week (or other time period) for each activity in the network. If crash costs are linear over time, the following formula can be used:

$$\text{Crash cost per period} = \frac{(\text{Crash cost} - \text{Normal cost})}{(\text{Normal time} - \text{Crash time})} (3-11)$$

STEP 2: Using the current activity times, find the critical path(s) in the project network. Identify the critical activities.

STEP 3: If there is only one critical path, then select the activity on this critical path that (a) can still be crashed and (b) has the smallest crash cost per period. Crash this activity by one period.

If there is more than one critical path, then select one activity from each critical path such that (a) each selected activity can still be crashed and (b) the total crash cost per period of *all* selected activities is the smallest. Crash each activity by one period. Note that the same activity may be common to more than one critical path.

STEP 4: Update all activity times. If the desired due date has been reached, stop. If not, return to Step 2.

We illustrate project crashing in Example 12.

Example 12

PROJECT CRASHING TO MEET A DEADLINE AT MILWAUKEE PAPER

Suppose the plant manager at Milwaukee Paper Manufacturing has been given only 13 weeks (instead of 16 weeks) to install the new pollution control equipment. As you recall, the length of Julie Ann Williams's critical path was 15 weeks, but she must now complete the project in 13 weeks.

APPROACH► Williams needs to determine which activities to crash, and by how much, to meet this 13-week due date. Naturally, Williams is interested in speeding up the project by 2 weeks, at the least additional cost.

SOLUTION► The company's normal and crash times, and normal and crash costs, are shown in Table 3.5. Note, for example, that activity B's normal time is 3 weeks (the estimate used in computing the critical path), and its crash time is 1 week. This means that activity B can be shortened by up to 2 weeks if extra resources are provided. The cost of these additional resources is \$4,000 (= difference between the crash cost of \$34,000 and the normal cost of \$30,000). If we assume that the crashing cost is linear over time (i.e., the cost is the same each week), activity B's crash cost per week is \$2,000 (= \$4,000/2).

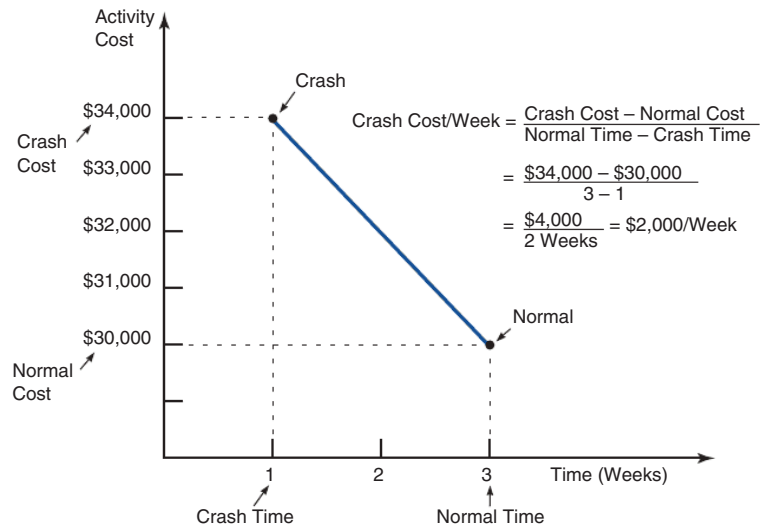
TABLE 3.5 Normal and Crash Data for Milwaukee Paper Manufacturing

ACTIVITY	TIME (WEEKS)		COST (\$)		CRASH COST PER WEEK (\$)	CRITICAL PATH?
	NORMAL	CRASH	NORMAL	CRASH		
A	2	1	22,000	22,750	750	Yes
B	3	1	30,000	34,000	2,000	No
C	2	1	26,000	27,000	1,000	Yes
D	4	3	48,000	49,000	1,000	No
E	4	2	56,000	58,000	1,000	Yes
F	3	2	30,000	30,500	500	No
G	5	2	80,000	84,500	1,500	Yes
H	2	1	16,000	19,000	3,000	Yes

This calculation for Activity B is shown in Figure 3.15. Crash costs for all other activities can be computed in a similar fashion.

Figure 3.15

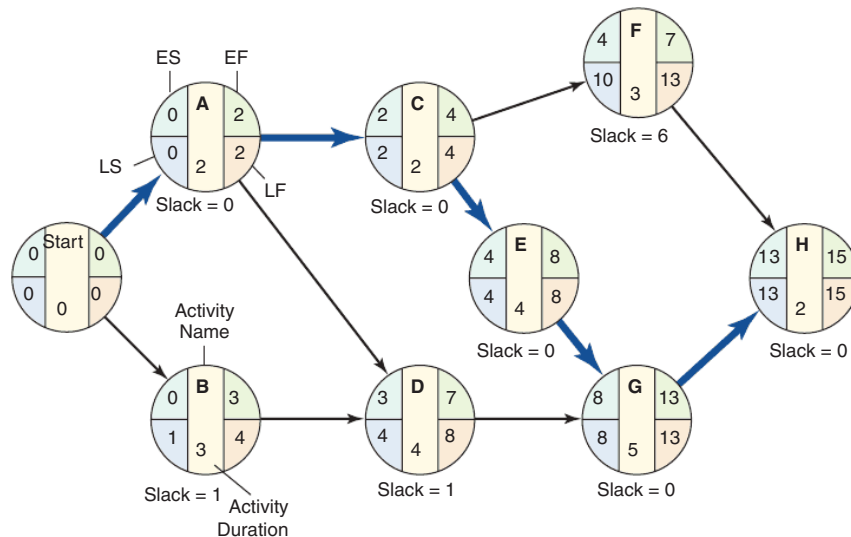
Crash and Normal Times and Costs for Activity B



Steps 2, 3, and 4 can now be applied to reduce Milwaukee Paper's project completion time at a minimum cost. We show the project network for Milwaukee Paper again in Figure 3.16.

Figure 3.16

Critical Path and Slack Times for Milwaukee Paper



The current critical path (using normal times) is Start–A–C–E–G–H, in which Start is just a dummy starting activity. Of these critical activities, activity A has the lowest crash cost per week of \$750. Julie Ann Williams should therefore crash activity A by 1 week to reduce the project completion time to 14 weeks. The cost is an additional \$750. Note that activity A cannot be crashed any further, since it has reached its crash limit of 1 week.

At this stage, the original path Start–A–C–E–G–H remains critical with a completion time of 14 weeks. However, a new path Start–B–D–G–H is also critical now, with a completion time of 14 weeks. Hence, any further crashing must be done to both critical paths.

On each of these critical paths, we need to identify one activity that can still be crashed. We also want the total cost of crashing an activity on each path to be the smallest. We might be tempted to simply pick the activities with the smallest crash cost per period in each path. If we did this, we would select activity C from the first path and activity D from the second path. The total crash cost would then be \$2,000 (= \$1,000 + \$1,000).

But we spot that activity G is common to both paths. That is, by crashing activity G, we will simultaneously reduce the completion time of both paths. Even though the \$1,500 crash cost for activity G is higher than that for activities C and D, we would still prefer crashing G because the total crashing cost will now be only \$1,500 (compared with the \$2,000 if we crash C and D).

INSIGHT ► To crash the project down to 13 weeks, Williams should crash activity A by 1 week and activity G by 1 week. The total additional cost will be \$2,250 ($= \$750 + \$1,500$). This is important because many contracts for projects include bonuses or penalties for early or late finishes.

LEARNING EXERCISE ► Say the crash cost for activity B is \$31,000 instead of \$34,000. How does this change the answer? [Answer: no change.]

RELATED PROBLEMS ► 3.28–3.33

EXCEL OM Data File Ch03Ex12.xls can be found in [MyLab Operations Management](#).

A Critique of PERT and CPM

As a critique of our discussions of PERT, here are some of its features about which operations managers need to be aware:

Advantages

1. Especially useful when scheduling and controlling large projects.
2. Straightforward concept and not mathematically complex.
3. Graphical networks help highlight relationships among project activities.
4. Critical path and slack time analyses help pinpoint activities that need to be closely watched.
5. Project documentation and graphs point out who is responsible for various activities.
6. Applicable to a wide variety of projects.
7. Useful in monitoring not only schedules but costs as well.

OM in Action

Behind the Tour de France

A Tour de France racing team needs the same large behind-the-scenes support that might be seen in the better-known Formula One or NBA playoff competitions. “A Tour de France team is like a large traveling circus,” says the coach of the Belkin team. “The public only sees the riders, but they could not function without the unseen support staff.”

The project management skills needed mean scheduling everyone from media relations staff to doctors to mechanics to the drivers of the team's huge truck, vans, and convoy of support cars and coach. Each move is made to guarantee that riders are in peak physical, nutritional, and psychological condition. Here are just some of the supplies the project management team for Belkin handles:

- ◆ 11 mattresses
- ◆ 36 aero suits, 45 bib shorts, 54 race jerseys, 250 podium caps
- ◆ 63 bikes
- ◆ 140 wheels, 220 tires
- ◆ 250 feeding bags, 3,000 water bottles
- ◆ 2,190 nutrition gels, 3,800 nutrition bars
- ◆ 10 jars of peanut butter, 10 boxes of chocolate sprinkles, 20 bags of wine gums, 20 jars of jam
- ◆ 80 kg of nuts, raisins, apricots, and figs, plus 50 kg of cereals



Marc Pagani Photography/Shutterstock

The project management behind a world-tour team is complex: These top teams often compete in two to three races simultaneously, in different countries and sometimes on different continents. Each team has 25–35 riders (9 compete in any single race), coming from different parts of the world, going to different races at different times, each with his own physique and strengths. They have customized bikes, uniforms, and food preferences. The support staff can include another 30 people.

Sources: [ComputerWeekly.com](#) (July, 2017); [BBC News](#) (July 6, 2014); and [ProjectManagement.com](#) (October 7, 2016).

Limitations

1. Project activities have to be clearly defined, independent, and stable in their relationships.
2. Precedence relationships must be specified and networked together.
3. Time estimates tend to be subjective and are subject to fudging by managers who fear the dangers of being overly optimistic or not pessimistic enough.
4. There is the inherent danger of placing too much emphasis on the longest, or critical, path. Near-critical paths need to be monitored closely as well.

Using Microsoft Project to Manage Projects

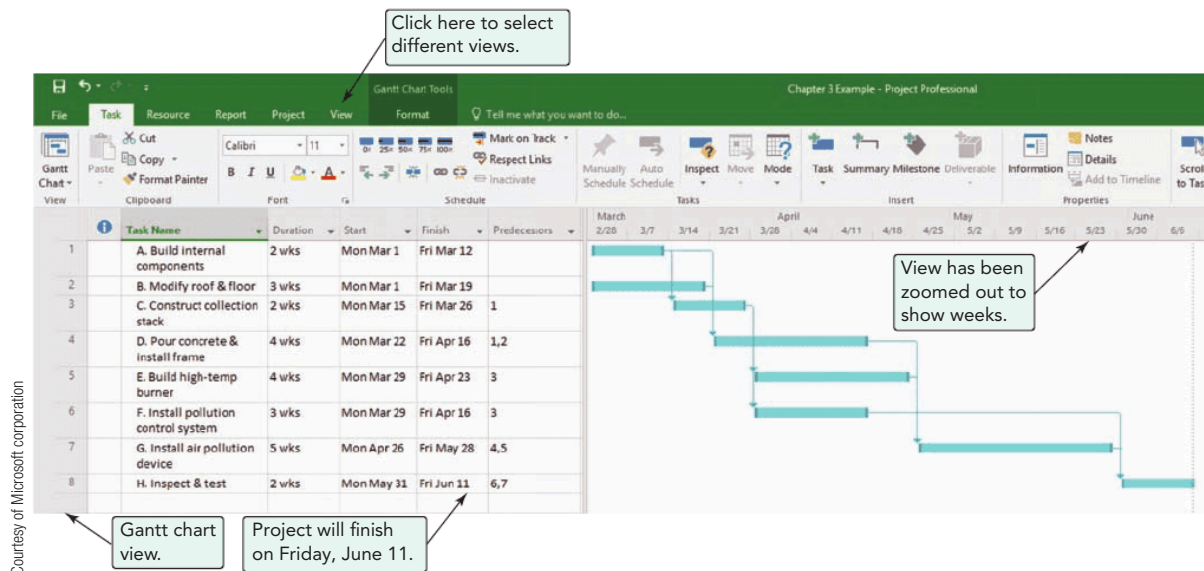
The approaches discussed so far are effective for managing small projects. However, for large or complex projects, specialized project management software is much preferred. In this section, we provide a brief introduction to the most popular example of such specialized software, Microsoft Project.

Microsoft Project is extremely useful in drawing project networks, identifying the project schedule, and managing project costs and other resources.

Entering Data Let us again consider the Milwaukee Paper Manufacturing project. Recall that this project has eight activities (repeated in the margin). The first step is to define the activities and their precedence relationships. To do so, we select **File|New** to open a blank project. We type the project start date (as March 1), then enter all activity information (see Program 3.1). For each activity (or task, as Microsoft Project calls it), we fill in the name and duration. The description of the activity is also placed in the **Task Name** column in Program 3.1. As we enter activities and durations, the software automatically inserts start and finish dates.

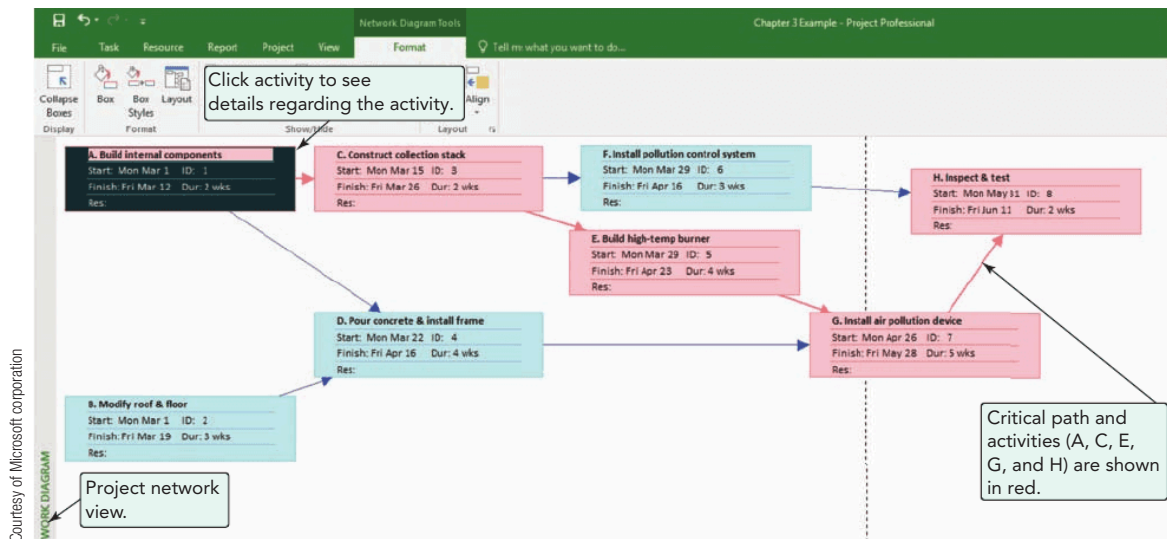
The next step is to define precedence relationships between these activities. To do so, we enter the relevant activity numbers (e.g., 1, 2) in the **Predecessors** column.

Milwaukee Paper Co. Activities		
ACTIVITY	TIME (WKS)	PREDE- CESSORS
A	2	—
B	3	—
C	2	A
D	4	A, B
E	4	C
F	3	C
G	5	D, E
H	2	F, G



Program 3.1

Gantt Chart in Microsoft Project for Milwaukee Paper Manufacturing



Program 3.2

Project Network in Microsoft Project for Milwaukee Paper Manufacturing

Viewing the Project Schedule When all links have been defined, the complete project schedule can be viewed as a Gantt chart. We can also select **View/Network Diagram** to view the schedule as a project network (shown in Program 3.2). The critical path is shown in red on the screen in the network diagram. We can click on any of the activities in the project network to view details of the activities. Likewise, we can easily add or remove activities from the project network. Each time we do so, Microsoft Project automatically updates all start dates, finish dates, and the critical path(s). If desired, we can manually change the layout of the network (e.g., reposition activities) by changing the options in **Format/Layout**.

Programs 3.1 and 3.2 show that if Milwaukee Paper's project starts March 1, it can be finished on June 11. The start and finish dates for all activities are also clearly identified. Project management software, we see, can greatly simplify the scheduling procedures discussed earlier in this chapter.

STUDENT TIP

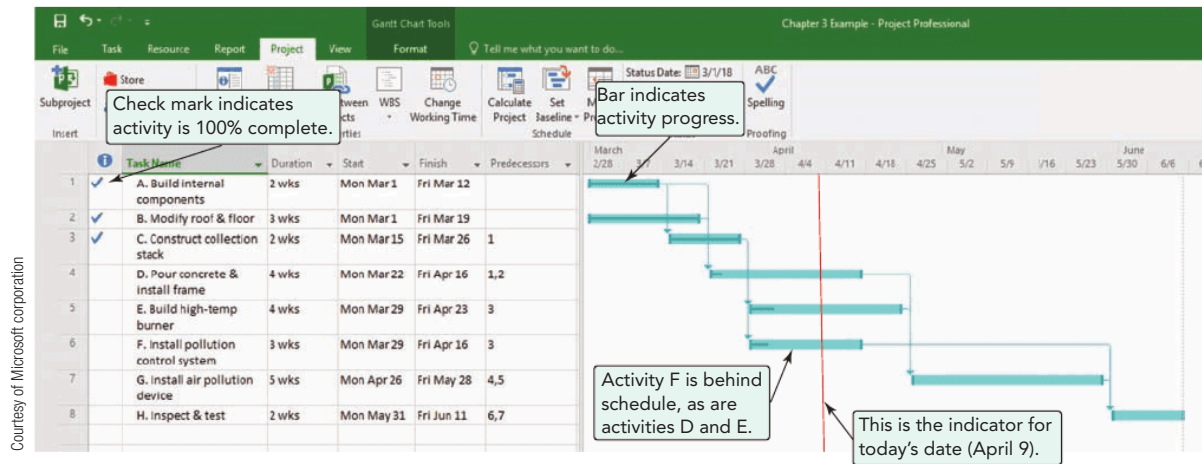
Now that you understand the workings of PERT and CPM, you are ready to master this useful program. Knowing such software gives you an edge over others in the job market.

Tracking the Time Status of a Project Perhaps the biggest advantage of using software to manage projects is that it can track the progress of the project. In this regard, Microsoft Project has many features available to track individual activities in terms of time, cost, resource usage, and so on.

An easy way to track the time progress of tasks is to enter the percent of work completed for each task. One way to do so is to double-click on any activity in the *Task Name* column in Program 3.1. A window is displayed that allows us to enter the percentage of work completed for each task.

The table in the margin provides data regarding the percentage of each of Milwaukee Paper's activities that are completed as of today. (Assume that today is Friday, April 9, i.e., the end of the sixth week of the project schedule.)

Pollution Project Percentage Completed on Apr. 9	
ACTIVITY	COMPLETED
A	100
B	100
C	100
D	10
E	20
F	20
G	0
H	0



Program 3.3

Tracking Project Progress in Microsoft Project

As shown in Program 3.3, the Gantt chart immediately reflects this updated information by drawing a thick line within each activity's bar. The length of this line is proportional to the percent of that activity's work that has been completed.

How do we know if we are on schedule? Notice that there is a vertical line shown on the Gantt chart corresponding to today's date. Microsoft Project will automatically move this line to correspond with the current date. If the project is on schedule, we should see all bars to the *left* of today's line indicate that they have been completed. For example, Program 3.3 shows that activities A, B, and C are on schedule. In contrast, activities D, E, and F appear to be behind schedule. These activities need to be investigated further to determine the reason for the delay. This type of easy *visual* information is what makes such software so useful in practice for project management.

Summary

PERT, CPM, and other scheduling techniques have proven to be valuable tools in controlling large and complex projects. Managers use such techniques to segment projects into discrete activities (work breakdown structures), identifying specific resources and time requirements for each. With PERT and CPM, managers can understand the status of each activity, including its earliest start, latest start, earliest finish, and latest finish (ES, LS, EF, and LF) times. By controlling the trade-off between ES and LS, managers can identify the activities that have slack and can address resource allocation, perhaps by smoothing resources. Effective project management also allows managers to focus on the activities that are critical to timely project

completion. By understanding the project's critical path, they know where crashing makes the most economic sense.

Good project management also allows firms to efficiently create products and services for global markets and to respond effectively to global competition. Microsoft Project, illustrated in this chapter, is one of a wide variety of software packages available to help managers handle network modeling problems.

The models described in this chapter require good management practices, detailed work breakdown structures, clear responsibilities assigned to activities, and straightforward and timely reporting systems. All are critical parts of project management.

Key Terms

Project organization (p. 94)	Critical path method (CPM) (p. 99)	Backward pass (p. 106)
Work breakdown structure (WBS) (p. 96)	Critical path (p. 99)	Slack time (p. 107)
Gantt charts (p. 97)	Activity-on-node (AON) (p. 100)	Optimistic time (p. 109)
Waterfall projects (p. 99)	Activity-on-arrow (AOA) (p. 100)	Pessimistic time (p. 109)
Agile projects (p. 99)	Dummy activity (p. 102)	Most likely time (p. 109)
Program evaluation and review technique (PERT) (p. 99)	Critical path analysis (p. 103)	Crashing (p. 114)
	Forward pass (p. 104)	

Ethical Dilemma

Two examples of massively mismanaged projects are TAURUS and the “Big Dig.” The first, formally called the London Stock Exchange Automation Project, cost \$575 million before it was finally abandoned. Although most IT projects have a reputation for cost overruns, delays, and underperformance, TAURUS set a new standard.

But even TAURUS paled next to the biggest, most expensive public works project in U.S. history—Boston’s 15-year-long Central Artery/Tunnel Project. Called the Big Dig, this was

perhaps the poorest and most felonious case of project mismanagement in decades. From a starting \$2 billion budget to a final price tag of \$15 billion, the Big Dig cost more than the Panama Canal, Hoover Dam, or Interstate 95, the 1,919-mile highway between Maine and Florida.

Read about one of these two projects (or another of your choice) and explain why it faced such problems. How and why do project managers allow such massive endeavors to fall into such a state? What do you think are the causes?

Discussion Questions

1. Give an example of a situation in which project management is needed.
2. Explain the purpose of project organization.
3. What are the three phases involved in the management of a large project?
4. What are some of the questions that can be answered with PERT and CPM?
5. What is a project and why is it important to understand the project environment?
6. What is the use of Gantt charts in project management?
7. What is the difference between an activity-on-arrow (AOA) network and an activity-on-node (AON) network? Which is primarily used in this chapter?
8. What are the ethical dilemmas that project managers are faced with when dealing with projects?
9. While numerous trade-offs may be required during the life cycle of a project, the trade-offs between cost, time, and performance are considered essential. Discuss.
10. Describe how expected activity times and variances can be computed in a PERT network.
11. Define *earliest start*, *earliest finish*, *latest finish*, and *latest start* times.
12. Students are sometimes confused by the concept of critical path, and want to believe that it is the *shortest* path through a network. Convincingly explain why this is not so.
13. What are dummy activities? Why are they used in activity-on-arrow (AOA) project networks?
14. What are the three time estimates used with PERT?
15. Would a project manager ever consider crashing a noncritical activity in a project network? Explain convincingly.
16. How is the variance of the total project computed in PERT?
17. Describe the meaning of slack, and discuss how it can be determined.
18. How can we determine the probability that a project will be completed by a certain date? What assumptions are made in this computation?
19. Compare and contrast the Project Evaluation and Review Technique (PERT) with the Critical Path Method (CPM).
20. What is the difference between the *waterfall* approach and *agile* project management?

Using Software to Solve Project Management Problems

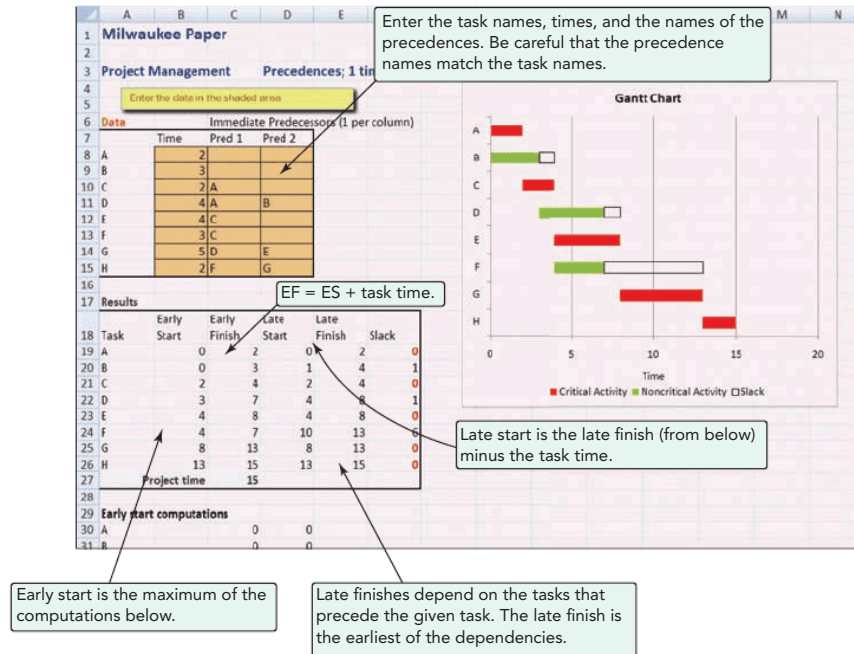
In addition to the Microsoft Project software illustrated earlier, both Excel OM and POM for Windows are available to readers of this text as project management tools in [MyLab Operations Management](#).

USING EXCEL OM

Excel OM has a Project Scheduling module. Program 3.4 uses the data from the Milwaukee Paper Manufacturing example in this chapter (see Examples 4 and 5). The PERT/CPM analysis also handles activities with three time estimates.

Program 3.4

Excel OM's Use of Milwaukee Paper Manufacturing's Data from Examples 4 and 5



USING POM FOR WINDOWS

POM for Windows' Project Scheduling module can also find the expected project completion time for a CPM and PERT network with either one or three time estimates. POM for Windows also performs project crashing. For further details refer to Appendix II.

Solved Problems

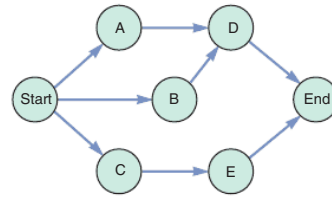
Virtual Office Hours help is available in [MyLab Operations Management](#).

SOLVED PROBLEM 3.1

Construct an AON network based on the following:

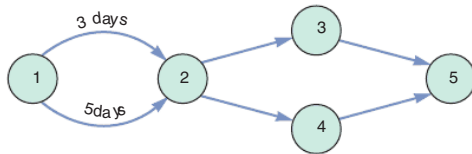
ACTIVITY	IMMEDIATE PREDECESSOR(S)
A	—
B	—
C	—
D	A, B
E	C

SOLUTION



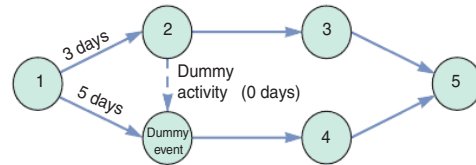
SOLVED PROBLEM 3.2

Insert a dummy activity and event to correct the following AOA network:



SOLUTION

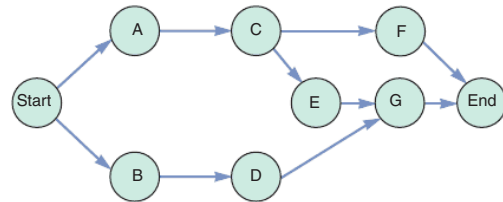
Because we cannot have two activities starting and ending at the same node, we add the following dummy activity and dummy event to obtain the correct AOA network:



SOLVED PROBLEM 3.3

Calculate the critical path, project completion time T , and project variance σ_p^2 based on the following AON network information:

ACTIVITY	TIME	VARIANCE	ES	EF	LS	LF	SLACK
A	2	$\frac{2}{6}$	0	2	0	2	0
B	3	$\frac{2}{6}$	0	3	1	4	1
C	2	$\frac{4}{6}$	2	4	2	4	0
D	4	$\frac{4}{6}$	3	7	4	8	1
E	4	$\frac{2}{6}$	4	8	4	8	0
F	3	$\frac{1}{6}$	4	7	10	13	6
G	5	$\frac{1}{6}$	8	13	8	13	0



SOLUTION

We conclude that the critical path is Start–A–C–E–G–End:

Total project time = $T = 2 + 2 + 4 + 5 = 13$
and

$$\sigma_p^2 = \sum \text{Variances on the critical path}$$

$$= \frac{2}{6} + \frac{4}{6} + \frac{2}{6} + \frac{1}{6} = \frac{9}{6} = 1.5$$

SOLVED PROBLEM 3.4

To complete the wing assembly for an experimental aircraft, Jim Gilbert has laid out the seven major activities involved. These activities have been labeled A through G in the following table, which also shows their estimated completion times (in weeks) and immediate predecessors. Determine the expected time and variance for each activity.

ACTIVITY	<i>a</i>	<i>m</i>	<i>b</i>	IMMEDIATE PREDECESSORS
A	1	2	3	—
B	2	3	4	—
C	4	5	6	A
D	8	9	10	B
E	2	5	8	C, D
F	4	5	6	D
G	1	2	3	E

SOLUTION

Expected times and variances can be computed using Equations (3-6) and (3-7) presented on page 110 in this chapter. The results are summarized in the following table:

ACTIVITY	EXPECTED TIME (IN WEEKS)	VARIANCE
A	2	$\frac{1}{9}$
B	3	$\frac{1}{9}$
C	5	$\frac{1}{9}$
D	9	$\frac{1}{9}$
E	5	1
F	5	$\frac{1}{9}$
G	2	$\frac{1}{9}$

SOLVED PROBLEM 3.5

Referring to Solved Problem 3.4, now Jim Gilbert would like to determine the critical path for the entire wing assembly project as well as the expected completion time for the total project. In addition, he would like to determine the earliest and latest start and finish times for all activities.

SOLUTION

The AON network for Gilbert's project is shown in Figure 3.17. Note that this project has multiple activities (A and B) with no immediate predecessors, and multiple activities (F and G) with no successors. Hence, in addition to a unique starting activity (Start), we have included a unique finishing activity (End) for the project.

Figure 3.17 shows the earliest and latest times for all activities. The results are also summarized in the following table:

ACTIVITY	ACTIVITY TIME				SLACK
	ES	EF	LS	LF	
A	0	2	5	7	5
B	0	3	0	3	0
C	2	7	7	12	5
D	3	12	3	12	0
E	12	17	12	17	0
F	12	17	14	19	2
G	17	19	17	19	0

Expected project length = 19 weeks

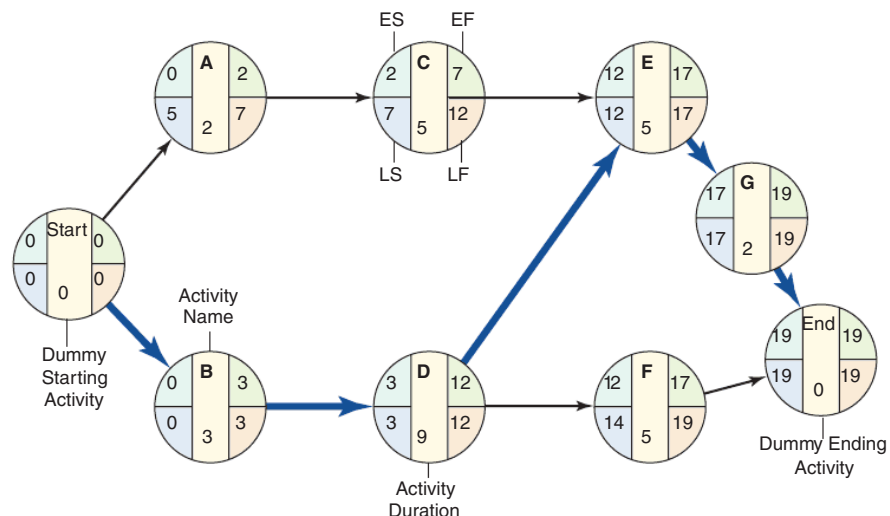
Variance of the critical path = 1.333

Standard deviation of the critical path = 1.155 weeks

The activities along the critical path are B, D, E, and G. These activities have zero slack as shown in the table.

Figure 3.17

Critical Path for Solved Problem 3.5



SOLVED PROBLEM 3.6

The following information has been computed from a project:

Expected total project time = $T = 62$ weeks

Project variance (σ_p^2) = 81

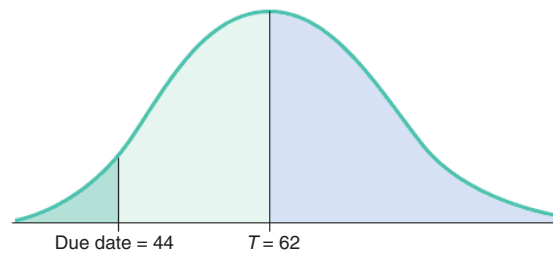
What is the probability that the project will be completed 18 weeks *before* its expected completion date?

SOLUTION

The desired completion date is 18 weeks before the expected completion date, 62 weeks. The desired completion date is 44 (or 62–18) weeks:

$$\begin{aligned}\sigma_p &= \sqrt{\text{Project variance}} \\ Z &= \frac{\text{Due date} - \text{Expected completion date}}{\sigma_p} \\ &= \frac{44 - 62}{9} = \frac{-18}{9} = -2.0\end{aligned}$$

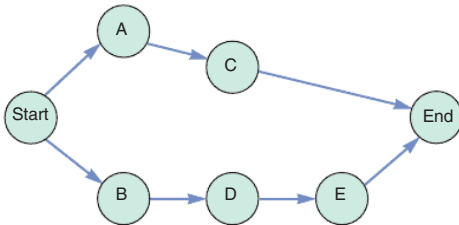
The normal curve appears as follows:



Because the normal curve is symmetrical and table values are calculated for positive values of Z , the area desired is equal to $1 - (\text{table value})$. For $Z = +2.0$ the area from the table is .97725. Thus, the area corresponding to a Z -value of -2.0 is .02275 (or $1 - .97725$). Hence, the probability of completing the project 18 weeks before the expected completion date is approximately .023, or 2.3%.

SOLVED PROBLEM 3.7

Determine the least cost of reducing the project completion date by 3 months based on the following information:



ACTIVITY	NORMAL TIME (MONTHS)	CRASH TIME (MONTHS)	NORMAL COST	CRASH COST
A	6	4	\$2,000	\$2,400
B	7	5	3,000	3,500
C	7	6	1,000	1,300
D	6	4	2,000	2,600
E	9	8	8,800	9,000

Next, crash cost/month must be computed for each activity:

ACTIVITY	NORMAL TIME-CRASH TIME	CRASH COST-NORMAL COST	CRASH COST/MONTH	CRITICAL PATH?
A	2	\$400	\$200/month	No
B	2	500	250/month	Yes
C	1	300	300/month	No
D	2	600	300/month	Yes
E	1	200	200/month	Yes

Finally, we will select that activity on the critical path with the smallest crash cost/month. This is activity E. Thus, we can reduce the total project completion date by 1 month for an additional cost of \$200. We still need to reduce the project completion date by 2 more months. This reduction can be achieved at least cost along the critical path by reducing activity B by 2 months for an additional cost of \$500. Neither reduction has an effect on noncritical activities. This solution is summarized in the following table:

ACTIVITY	MONTHS REDUCED	COST
E	1	\$200
B	2	500
		Total: \$700

SOLUTION

The first step in this problem is to compute ES, EF, LS, LF, and slack for each activity.

ACTIVITY	ES	EF	LS	LF	SLACK
A	0	6	9	15	9
B	0	7	0	7	0
C	6	13	15	22	9
D	7	13	7	13	0
E	13	22	13	22	0

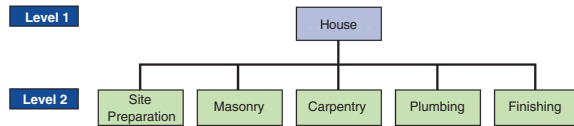
The critical path consists of activities B, D, and E.

Problems

Note: **Px** means the problem may be solved with POM for Windows and/or Excel OM.

Problems 3.1–3.2 relate to Project Planning

- **3.1** The work breakdown structure (WBS) for building a house (levels 1 and 2) is shown below:



- Add two level-3 activities to each of the level-2 activities to provide more detail to the WBS.
- Select one of your level-3 activities and add two level-4 activities below it.

- • **3.2** You are the project manager for a project in which you are going to organize a conference. Develop a work breakdown structure (WBS) in the form of a hierarchical framework, similar to the one in Figure 3.3, based on the partial list shown in the table with all major deliverables, component deliverables, and component parts.

LEVEL	ACTIVITY
10	Conference
11	Sessions
11.01	Papers
11.01.01	Topic A
11.01.02	Topic B
11.02	Panels
11.02.01	Panel A
12	Location & Dates
12.01	Site X
12.02	Site Y
12.03	Date A
12.04	Date B
13	Facilities
13.01	Buildings
13.01.01	Building East
13.01.02	Building West
13.02	Equipment
13.03	Food
13.03.01	Breakfast
13.03.02	Coffee

Problem 3.3 relates to Project Scheduling

- • **3.3** Potamia Construction Co. (PCC) is developing plans to erect a new hospital building. Potamia Construction Co. has established the project activities and their immediate predecessor activities together with their estimated activity durations according to the information described in the following table. Prepare a Gantt chart (horizontal bar chart) for the whole construction activity based on this information.

CODE	ACTIVITY	IMMEDIATE PREDECESSOR(S)	DURATION (DAYS)
A	Buy the land	None	40
B	Get the construction permission	A	30
C	Excavate	B	10
D	Erect building	C	100
E	Electrical work	D	10
F	Plumbing work	D	20
G	Paint interior	D	10
H	Clean up building	E, F, G	5
I	Move in furniture	H	5

Problems 3.4–3.14 relate to Project Management Techniques

- • **3.4** Using the project information listed in the table below:
 - Draw the AON network for the construction activity.
 - Draw the AOA for the construction activity.

ACTIVITY	IMMEDIATE PREDECESSOR(S)	DURATION (DAYS)
A	—	4
B	—	2
C	—	2
D	A	2
E	B	3
F	B	4
G	C	2
H	D, E	5
I	F, G	3

- **3.5** The table below shows the activities, immediate predecessors, and activity durations of a project. Draw the AON project network. How long would it take to complete this project? What is the length of the critical path?

ACTIVITY	IMMEDIATE PREDECESSOR(S)	DURATION (DAYS)
A	—	2
B	A	4
C	A	4
D	B	6
E	D	9
F	C	8
G	E, F	2
H	G	2
I	G	3
J	H, I	4
K	J	4

- **3.6** A project has the following activities, precedence relationships, and activity durations. Draw the AOA project network. How long would it take to complete this project? What is the length of the critical path?

ACTIVITY	IMMEDIATE PREDECESSOR(S)	DURATION (WEEKS)
A	None	18
B	A	8
C	B	6
D	A	9
E	C, D	5
F	E	4
G	D	10
H	E	11
I	G, H	3

- **3.7** Draw the AON project network using the following information:

ACTIVITY	IMMEDIATE PREDECESSOR(S)	DURATION (WEEKS)
A	—	4
B	—	35
C	—	5
D	A	9
E	B	3
F	C, E	1
G	D	2
H	F	3
I	G, H	15
J	G	1

How long would it take to complete this project? Which activities are included on the critical path?

- **3.8** Roger Ginde is developing a program in supply chain management certification for managers. Ginde has listed a number of activities that must be completed before a training program of this nature could be conducted. The activities, immediate predecessors, and times appear in the accompanying table:

ACTIVITY	IMMEDIATE PREDECESSOR(S)	TIME (DAYS)
A	—	2
B	—	5
C	—	1
D	B	10
E	A, D	3
F	C	6
G	E, F	8

- Develop an AON network for this problem.
- What is the critical path?
- What is the total project completion time?
- What is the slack time for each individual activity? 

- **3.9** Task time estimates for the modification of an assembly line at Jim Goodale's Carbondale, Illinois, factory are as follows:

ACTIVITY	TIME (IN HOURS)	IMMEDIATE PREDECESSORS
A	6.0	—
B	7.2	—
C	5.0	A
D	6.0	B, C
E	4.5	B, C
F	7.7	D
G	4.0	E, F

- Draw the project network using AON.
- Identify the critical path.
- What is the expected project completion time?
- Draw a Gantt chart for the project. 

- **3.10** The activities described by the following table are given for the Howard Corporation in Kansas:

ACTIVITY	IMMEDIATE PREDECESSOR(S)	TIME
A	—	9
B	A	7
C	A	3
D	B	6
E	B	9
F	C	4
G	E, F	6
H	D	5
I	G, H	3

- Draw the appropriate AON PERT diagram for J.C. Howard's management team.
- Find the critical path.
- What is the project completion time? 

- **3.11** Draw the AON project network using the following information:

ACTIVITY	IMMEDIATE PREDECESSOR(S)	DURATION (WEEKS)
A	—	4
B	A	3
C	A	6
D	B	2
E	B	3
F	C, D	5
G	E, F	1

- Draw an AON diagram of the project, including activity durations.
- Define the critical path, listing all critical activities in chronological order.
- What is the project duration (in weeks)?
- What is the slack (in weeks) associated with any and all non-critical paths through the project?

- **3.12** The activities needed to build a prototype laser scanning machine at Dave Fletcher Corp. are listed in the following table. Construct an AON network for these activities.

ACTIVITY	IMMEDIATE PREDECESSOR(S)	ACTIVITY	IMMEDIATE PREDECESSOR(S)
A	—	E	B
B	—	F	B
C	A	G	C, E
D	A	H	D, F

ACTIVITY	TIME (WEEKS)	ACTIVITY	TIME (WEEKS)
A	6	E	4
B	7	F	6
C	3	G	10
D	2	H	7

Additional problem 3.16 is available in MyLab Operations Management. It involves 21 activities in building the Rover 6 auto.

- **3.13** The following represent activities in Marc Massoud's Construction Company project.

ACTIVITY	IMMEDIATE PREDECESSOR(S)	TIME (WEEKS)
A	—	1
B	—	4
C	A	1
D	B	5
E	B	2
F	C, E	7
G	D	2
H	F, G	3

- Draw the network to represent this situation.
- Which activities are on the critical path?
- What is the length of the critical path? **Px**

Additional problem 3.14 is available in MyLab Operations Management. It involves finding the activities for a college degree.

Problems 3.15–3.16 relate to Determining the Project Schedule

- **3.15** Dave Fletcher (see Problem 3.12) was able to determine the activity times for constructing his laser scanning machine. Fletcher would like to determine ES, EF, LS, LF, and slack for each activity. The total project completion time and the critical path should also be determined. Here are the activity times:

Problems 3.17–3.27 relate to Variability in Activity Times

- **3.17** Ross Hopkins, president of Hopkins Hospitality, has developed the tasks, durations, and predecessor relationships in the following table for building new motels. Draw the AON network and answer the questions that follow.

ACTIVITY	IMMEDIATE PREDECESSOR(S)	TIME ESTIMATES (IN WEEKS)		
		OPTIMISTIC	MOST LIKELY	PESSIMISTIC
A	—	4	8	10
B	A	2	8	24
C	A	8	12	16
D	A	4	6	10
E	B	1	2	3
F	E, C	6	8	20
G	E, C	2	3	4
H	F	2	2	2
I	F	6	6	6
J	D, G, H	4	6	12
K	I, J	2	2	3

- What is the expected (estimated) time for activity C?
- What is the variance for activity C?
- Based on the calculation of estimated times, what is the critical path?
- What is the estimated time of the critical path?
- What is the activity variance along the critical path?
- What is the probability of completion of the project before week 36? **Px**

- **3.18** A renovation of the gift shop at Orlando Amway Center has six activities (in hours). For the following estimates of a , m , and b , calculate the expected time and the standard deviation for each activity: **Px**

ACTIVITY	a	m	b
A	11	15	19
B	27	31	41
C	18	18	18
D	8	13	19
E	17	18	20
F	16	19	22

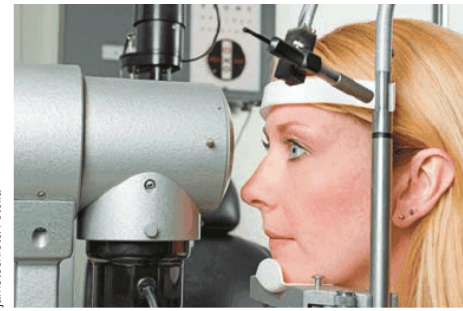
- **3.19** Kelle Carpet and Trim installs carpet in commercial offices. Peter Kelle has been very concerned with the amount of time it took to complete several recent jobs. Some of his workers are very unreliable. A list of activities and their optimistic completion time, the most likely completion time, and the pessimistic completion time (all in days) for a new contract are given in the following table:

ACTIVITY	TIME (DAYS)			IMMEDIATE PREDECESSOR(S)
	a	m	b	
A	3	6	8	—
B	2	4	4	—
C	1	2	3	—
D	6	7	8	C
E	2	4	6	B, D
F	6	10	14	A, E
G	1	2	4	A, E
H	3	6	9	F
I	10	11	12	G
J	14	16	20	C
K	2	8	10	H, I

- Determine the expected completion time and variance for each activity.
- Determine the total project completion time and the critical path for the project.
- Determine ES, EF, LS, LF, and slack for each activity.
- What is the probability that Kelle Carpet and Trim will finish the project in 40 days or less? **Px**

- **3.20** The estimated times and immediate predecessors for the activities in a project at George Kyparis's retinal scanning company are given in the following table. Assume that the activity times are independent.

ACTIVITY	IMMEDIATE PREDECESSOR	TIME (WEEKS)		
		a	m	b
A	—	9	10	11
B	—	4	10	16
C	A	9	10	11
D	B	5	8	11



- Calculate the expected time and variance for each activity.
- What is the expected completion time of the critical path? What is the expected completion time of the other path in the network?
- What is the variance of the critical path? What is the variance of the other path in the network?
- If the time to complete path A–C is normally distributed, what is the probability that this path will be finished in 22 weeks or less?
- If the time to complete path B–D is normally distributed, what is the probability that this path will be finished in 22 weeks or less?
- Explain why the probability that the *critical path* will be finished in 22 weeks or less is not necessarily the probability that the *project* will be finished in 22 weeks or less. **Px**

- **3.21** Coleman Rich Control Devices, Inc., produces custom-built relay devices for auto makers. The most recent project undertaken by Rich requires 14 different activities. Rich's managers would like to determine the total project completion time (in days) and those activities that lie along the critical path. The appropriate data are shown in the following table.

- What is the probability of being done in 53 days?
- What date results in a 99% probability of completion? **Px**

ACTIVITY	IMMEDIATE PREDECESSOR(S)	OPTIMISTIC TIME	MOST LIKELY TIME	PESSIMISTIC TIME
A	—	4	6	7
B	—	1	2	3
C	A	6	6	6
D	A	5	8	11
E	B, C	1	9	18
F	D	2	3	6
G	D	1	7	8
H	E, F	4	4	6
I	G, H	1	6	8
J	I	2	5	7
K	I	8	9	11
L	J	2	4	6
M	K	1	2	3
N	L, M	6	8	10

••• **3.22** Four Squares Productions, a firm hired to coordinate the release of the movie *Pirates of the Caribbean: Dead Men Tell No Tales* (starring Johnny Depp), identified 16 activities to be completed before the release of the film.

ACTIVITY	IMMEDIATE PREDECESSOR(S)	OPTIMISTIC TIME	MOST LIKELY TIME	PESSIMISTIC TIME
A	—	1	2	4
B	—	3	3.5	4
C	—	10	12	13
D	—	4	5	7
E	—	2	4	5
F	A	6	7	8
G	B	2	4	5.5
H	C	5	7.7	9
I	C	9.9	10	12
J	C	2	4	5
K	D	2	4	6
L	E	2	4	6
M	F, G, H	5	6	6.5
N	J, K, L	1	1.1	2
O	I, M	5	7	8
P	N	5	7	9

- How many weeks in advance of the film release should Four Squares have started its marketing campaign? What is the critical path? The tasks (in time units of weeks) are as follows:
- What is the probability of completing the marketing campaign in the time (in weeks) noted in part a?
- If activities I and J were not necessary, what impact would this have on the critical path and the number of weeks needed to complete the marketing campaign? **Px**

•• **3.23** Using PERT, Adam Munson was able to determine that the expected project completion time for the construction of a pleasure yacht is 21 months, and the project variance is 4.

- What is the probability that the project will be completed in 17 months?
- What is the probability that the project will be completed in 20 months?
- What is the probability that the project will be completed in 23 months?
- What is the probability that the project will be completed in 25 months?
- What is the due date that yields a 95% chance of completion? **Px**

•• **3.24** A small software development project at Krishna Dhir's firm has five major activities. The times are estimated and provided in the following table.

ACTIVITY	IMMEDIATE PREDECESSOR	<i>a</i>	<i>m</i>	<i>b</i>
A	—	2	5	8
B	—	3	6	9
C	A	4	7	10
D	B	2	5	14
E	C	3	3	3

- Find the expected completion time for this project.
- What variance value would be used to find probabilities of finishing by a certain time? **Px**

••• **3.25** Janice Eliasson must complete the activities in the following table to finish her consulting project.

ACTIVITY	EXPECTED TIME	STANDARD DEVIATION OF TIME ESTIMATE	IMMEDIATE PREDECESSOR(S)
A	7	2	—
B	3	1	A
C	9	3	A
D	4	1	B, C
E	5	1	B, C
F	8	2	E
G	8	1	D, F
H	6	2	G

- Draw the appropriate PERT diagram.
- Find the critical path and project completion time.
- Find the probability that the project will take more than 49 time periods to complete. **Px**

Additional problems 3.26–3.27 are available in MyLab Operations Management. These have 12 and 23 PERT activities, respectively.

Problems 3.28–3.33 relate to Cost-Time Trade-Offs and Project Crashing

•• **3.28** Given the project diagram that follows, see graph (a) and the details of the accompanying table. How can you reduce the completion time of the project by four weeks in the most economical way?

TASK	DURATION (WEEKS)	CRASH TIME	MAXIMUM WEEKS CRASHED	NORMAL COST (€)	CRASH COST (€)	CRASH COST (€)/WEEK
A	3	2	1	4,000	5,000	1000
B	5	5	0	10,000	10,000	0
C	2	1	1	1,500	2,000	500
D	2	1	1	6,000	10,000	4,000
E	3	2	1	8,000	15,000	7,000
F	4	3	1	3,000	4,000	1000
G	8	6	2	4,000	7,000	3,000
H	2	1	1	3,000	5,000	2,000